# William Cheung

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## SUMMARY

Senior Data Scientist with 5+ years of experience leading machine learning initiatives, experimentation frameworks, and large-scale data solutions in the digital advertising and media industries. Proven ability to drive business impact through A/B testing, algorithm optimization, and cross-functional collaboration with product, engineering, and analytics teams. Passionate about building data-driven strategies that enhance product performance, operational efficiency, and client success.

## EXPERIENCE

**Squarespace;** New York, N.Y. July 2025 – Present
**Senior Data Scientist**
**Marketing Science**
● Develop and maintain multi-touch attribution pipelines, including integrating a paid media partner's data (sensor, touchpoint definitions, weight computation) and incorporating A/B and geo-test results into incrementality-adjusted channel weights (Using BigQuery, dbt, Airflow, Python)
● Build and calibrate Marketing Mix Models across multiple geos and channels, including Hill function saturation curves and spend optimization, and evaluate Google Meridian for adoption (Using Python, NumPyro, SciPy)
● Productionize MMM outputs by building warehouse tables for model weights, contributions, and raw touchpoints, and shipping recurring weight updates that feed channel-level attribution (Using BigQuery, dbt)
● Prototype a Media Measurement Hub with scenario modeling, diminishing-returns curves, and upcoming-spend inputs so marketing stakeholders can plan budget allocation (Using Hex, Python)
● Improve LTV modeling and data quality through segment-level LTV computation, imputation fixes, and monitoring queries on subscription score tables, increasing trust in the LTV metric used by marketing and finance (Using BigQuery, dbt)
**Agent / LLM Evals**
● Lead evaluation design for an internal NL-to-SQL conversational analytics agent, combining an LLM-as-judge framework (custom rubric criteria) with deterministic SQL correctness checks and a golden benchmark dataset (Using DeepEval, Python, BigQuery)
● Built an NLG scoring service (FastAPI, 7 scoring endpoints, batch scoring) and packaged it as an installable library reused across agent projects (Using Python, FastAPI, Gemini)
● Implemented offline batch evals, online eval gates, and pre-call, pre-execute, and post-execute guardrails in shadow mode, plus a confidence score for generated queries (Using LangGraph, Airflow, BigQuery)

**Goodway Group;** New York, N.Y. April 2023 – July 2025
**Senior Data Scientist**

● Lead the improvement and further **development of media buying algorithms** focusing on pricing and volume control. Benchmarked media buying algorithms against Trade Desk Koa, achieving a 23% reduction in CPM without significant impact on CPA.
● Build internal **Marketing Mix Modeling tools as well as a Scenario Modeling** feature extension using open-source packages and facilitate model storage through MLflow. This lead to a 17-53% increase in ROAS for various clients.
● Collaborated with product, engineering, and client teams to **deploy internal web applications** for stakeholders with Streamlit (e.g. A/B framework app, Brand assessment app, Daido chatbot, etc.), with up to 45% usage rate among client teams.
● Create **processes for internal A/B testing** for both back-end optimization and guide the analytics teams on launching client-side tests.

**Criteo;** New York, N.Y. May 2021 – April 2023
***Data Scientist***
● Built self-service reporting and performance insights into online video advertising campaigns (OLV and OTT) in order to help **grow client spend ~300% year over year** on upper funnel and full funnel tactics
● Supported backend data pipelines and data migration, with dataflow job, and warehouse management for tasks such as **audience targeting**, complex **custom reporting**, and **supply data to other teams** for their analytics
● **Reviewed code** submissions on various database or automated tooling alterations
● **Built A/B testing infrastructure**, joining internal processes with a newly acquired platform (IponWeb) to allow internal teams to conduct A/B tests
● Used **Natural Language Processing to classify app inventory type** and allow for better reporting and training data for bidding engine

**Fordham University;** New York, N.Y September 2019 – May 2021
***Graduate Researcher (Machine Learning)***
● **Main Project - Using passive biometrics to continuously certify a user with multiple devices** (*Implicit continuous biometric authentication model using IoT wearables*) Initially testing with classical models (RF, KNN, SVM, NB) to artificial neural networks (CNN,RNN,CRNN) we've made a user authentication modeled for internet connected devices taking into account Heart Rate, Gait, and Breathing with accuracy of 97%. For implementation I've **built a full-stack system incorporating a custom Fitbit application front end to a computational back-end android app with server interaction** [JavaScript: Fitbit and server, java: android app, python: feature computation and model decision generation] of the authentication model
● **Secondary Project – User Location Prediction in Absence of** GPS (Point of Interest identification) study using machine learning with 95% accuracy

**National Grid;** Brooklyn, N.Y June 2016 – September 2019
***Financial Analyst***
● Automated system batch entries using VBA and SAP to **reduce manual entries by 15%** and speed up the monthly accounting process by an additional work day
● Analyzed financial information during month end close for the National Grid subsidiary, Narragansett Electric Company and work with business partners to accurately establish **cost capture methods for recovery of over $10 billion**

## EDUCATION

**Fordham University (**N.Y., N.Y.) Graduated - Dec. 2020
M.S in Data Science (GPA: 4.0)

**Stony Brook University (**Stony Brook, N.Y.) Graduated - May 2016
B.S. in Applied Mathematics and Economics [Double Major] (GPA: 3.6)

## Technical Skills

- **Programming & Scripting:** Python (Pandas, NumPy, Scikit-learn, TensorFlow), SQL, R, Bash
- **Machine Learning & Experimentation:** A/B Testing, Regression, Classification, NLP, Recommendation Systems, Time Series Forecasting, MLflow
- **Big Data & Cloud Technologies:** AWS, Google Cloud Platform, Presto
- **Development & Automation:** Git, Airflow, REST APIs
- Visualization & Tools: Streamlit, Tableau

## PUBLICATIONS

- Cheung, W and Vhaduri, S "Continuous Authentication of Wearable Device Users from Heart Rate, Gait, and Breathing Data" IEEE RAS/EMBS International Conference on Biomedical Robotics & Biomechatronics *https://arxiv.org/abs/2008.10779*
- Cheung, W and Vhaduri, S "Context-Dependent Implicit Authentication for Wearable Device User" IEEE International Symposium on Personal, Indoor and Mobile Radio Communications *https://arxiv.org/abs/2008.12145*